



Electronic I

Lecture 5

Diode Applications : Zener Diodes

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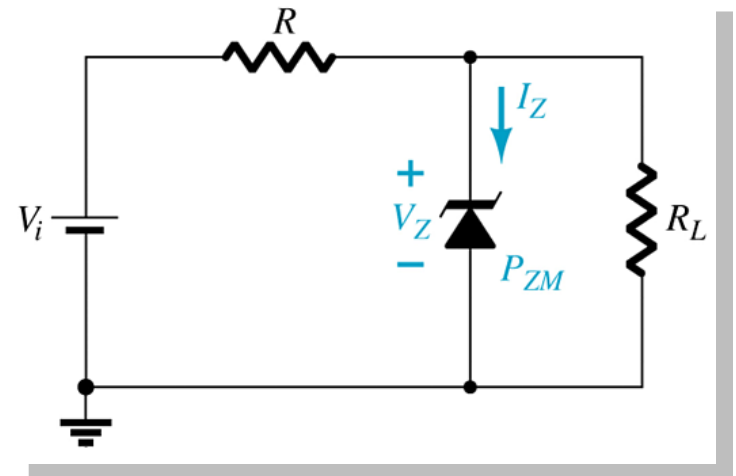
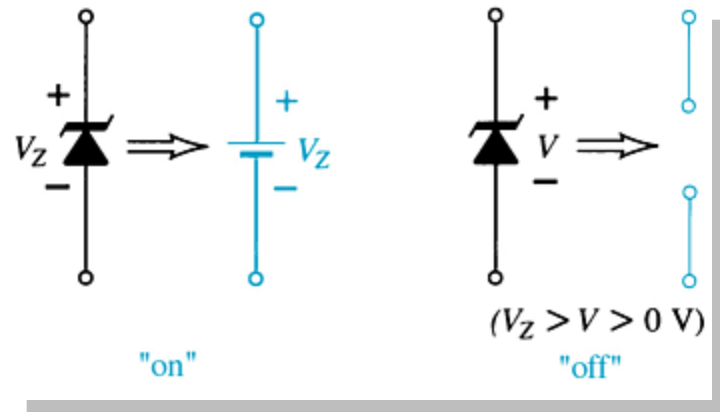
First Year / Spring Semester

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Zener Diodes

The Zener is a diode operated in reverse bias at the Zener Voltage (V_Z).

- When $V_i \geq V_Z$
 - The Zener is on
 - Voltage across the Zener is V_Z
 - Zener current: $I_Z = I_R - I_{RL}$
 - The Zener Power: $P_Z = V_Z I_Z$
- When $V_i < V_Z$
 - The Zener is off
 - The Zener acts as an open circuit



V_i and R

1. Determine the state of the Zener diode by removing it from the network and calculating the voltage across the resulting open circuit.

$$V = V_L = \frac{R_L V_i}{R + R_L}$$

2. Substitute the appropriate equivalent circuit and solve for the desired unknowns.

$$V_L = V_Z$$

$$I_Z = I_R - I_L$$

$$P_Z = V_Z I_Z$$

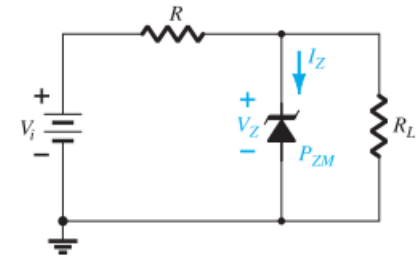


FIG. 2.112

Basic Zener regulator.

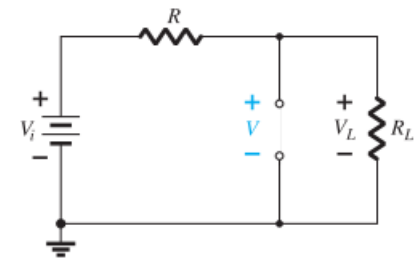


FIG. 2.113

Determining the state of the Zener diode.

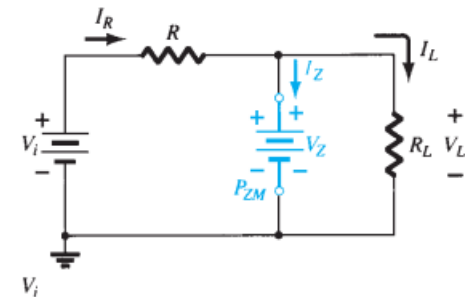
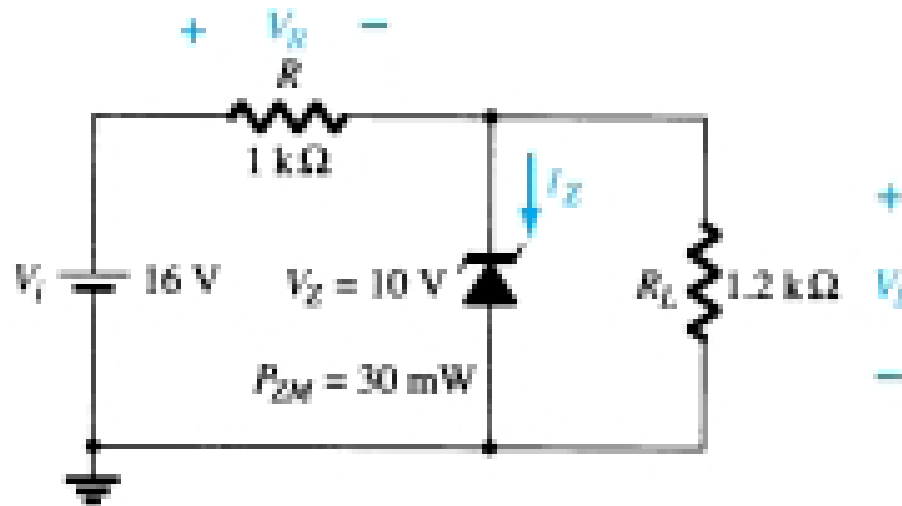


FIG. 2.114

Substituting the Zener equivalent for the "on" situation.

Example

For the following Zener diode network, determine V_L , V_R , I_Z , and P_Z .



Following the suggested procedure:

$$V = \frac{R_L V_I}{R + R_L} = \frac{1.2 \text{ k}\Omega (16 \text{ V})}{1 \text{ k}\Omega + 1.2 \text{ k}\Omega} = 8.73 \text{ V}$$

Since V_L is less than V_Z , the diode is in the “off” state.

Substituting the open-circuit equivalent will result in;

$$V_L = V = 8.73 \text{ V}$$

$$V_R = V_I - V_L = 16 \text{ V} - 8.73 \text{ V} = 7.27 \text{ V}$$

$$I_Z = 0 \text{ A}$$

$$P_Z = V_Z I_Z = V_Z (0 \text{ A}) = 0 \text{ W}$$

Fixed V_i , Variable R_L

- To determine the minimum load resistance that will turn the Zener diode on, simply calculate the value of R_L that will result in a load voltage $V_L = V_Z$.

That is,
$$V_L = V_Z = \frac{R_L V_i}{R_L + R}$$

- Solving for R_L , we have

$$R_{L_{\min}} = \frac{R V_Z}{V_i - V_Z}$$

$$I_{L_{\max}} = \frac{V_L}{R_L} = \frac{V_Z}{R_{L_{\min}}}$$

- Once the diode is in the “on” state, the voltage across R remains fixed at

$$V_R = V_i - V_Z$$

- and I_R remains fixed at

$$I_R = \frac{V_R}{R}$$

- **The Zener current**

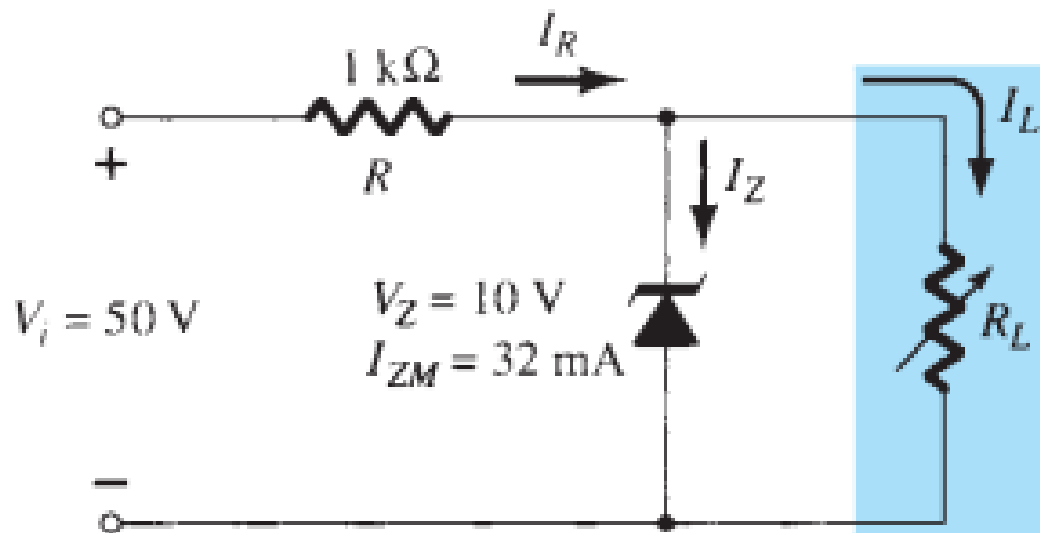
$$I_Z = I_R - I_L$$

$$I_{L_{\min}} = I_R - I_{ZM}$$

$$R_{L_{\max}} = \frac{V_Z}{I_{L_{\min}}}$$

EXAMPLE

- (a) For the following network , determine the range of R_L and I_L that will result in V_{RL} being maintained at 10 V.
- (b) Determine the maximum wattage rating of the diode.



(a) To determine the value of R_L that will turn the Zener diode on

$$R_{L_{\min}} = \frac{RV_Z}{V_I - V_Z} = \frac{(1 \text{ k}\Omega)(10 \text{ V})}{50 \text{ V} - 10 \text{ V}} = \frac{10 \text{ k}\Omega}{40} = \mathbf{250 \text{ }\Omega}$$

The voltage across the resistor R

$$V_R = V_I - V_Z = 50 \text{ V} - 10 \text{ V} = \mathbf{40 \text{ V}}$$

$$I_R = \frac{V_R}{R} = \frac{40 \text{ V}}{1 \text{ k}\Omega} = \mathbf{40 \text{ mA}}$$

$$I_{L_{\min}} = I_R - I_{ZM} = 40 \text{ mA} - 32 \text{ mA} = \mathbf{8 \text{ mA}}$$

$$R_{L_{\max}} = \frac{V_Z}{I_{L_{\min}}} = \frac{10 \text{ V}}{8 \text{ mA}} = \mathbf{1.25 \text{ k}\Omega}$$

$$\begin{aligned} \text{(b) } P_{\max} &= V_Z I_{ZM} \\ &= (10 \text{ V})(32 \text{ mA}) = \mathbf{320 \text{ mW}} \end{aligned}$$

Fixed R_L , Variable V_i

- For fixed values of R_L , the voltage V_i must be sufficiently large to turn the Zener diode on. The minimum turn-on voltage V_i $V_{i\min}$ is determined by

$$V_L = V_Z = \frac{R_L V_i}{R_L + R}$$

$$V_{i\min} = \frac{(R_L + R)V_Z}{R_L}$$

$$I_{R\max} = I_{ZM} + I_L$$

$$V_{i\max} = V_{R\max} + V_Z$$

$$V_{i\max} = I_{R\max}R + V_Z$$

EXAMPLE 2.28 Determine the range of values of V_i that will maintain the Zener diode of Fig. 2.121 in the “on” state.

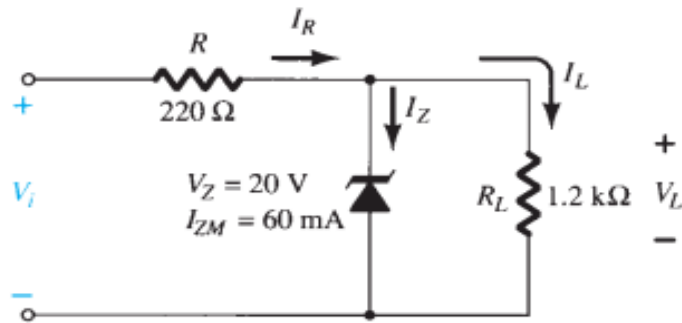


FIG. 2.121

$$V_{i_{\min}} = \frac{(R_L + R)V_Z}{R_L} = \frac{(1200\ \Omega + 220\ \Omega)(20\ \text{V})}{1200\ \Omega} = \mathbf{23.67\ \text{V}}$$

$$I_L = \frac{V_L}{R_L} = \frac{V_Z}{R_L} = \frac{20\ \text{V}}{1.2\ \text{k}\Omega} = 16.67\ \text{mA}$$

$$\begin{aligned} I_{R_{\max}} &= I_{ZM} + I_L = 60\ \text{mA} + 16.67\ \text{mA} \\ &= 76.67\ \text{mA} \end{aligned}$$

$$\begin{aligned} V_{i_{\max}} &= I_{R_{\max}}R + V_Z \\ &= (76.67\ \text{mA})(0.22\ \text{k}\Omega) + 20\ \text{V} \\ &= 16.87\ \text{V} + 20\ \text{V} \\ &= \mathbf{36.87\ \text{V}} \end{aligned}$$

AC Regulator and Square-Wave Generator

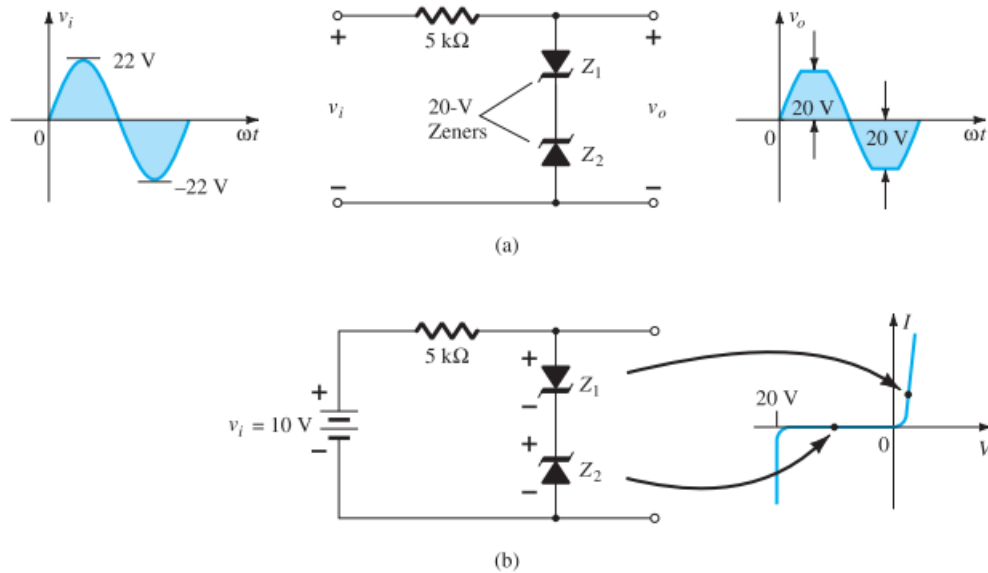


FIG. 2.144

Sinusoidal ac regulation: (a) 40-V peak-to-peak sinusoidal ac regulator; (b) circuit operation at $v_i = 10$ V.

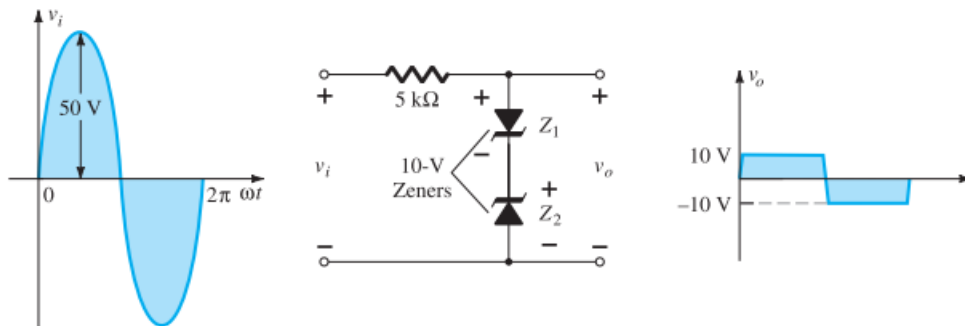
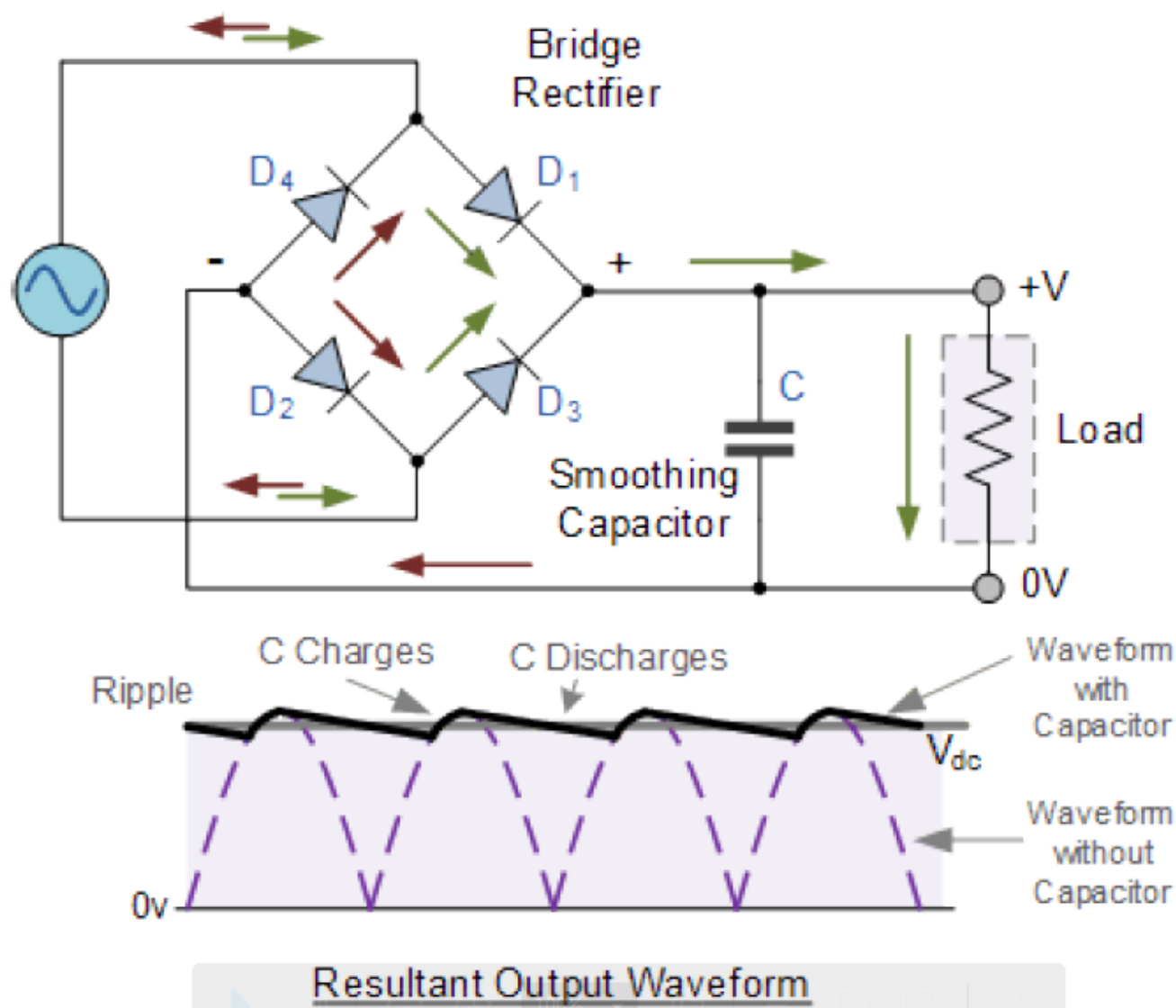


FIG. 2.145

Simple square-wave generator.

Full Wave Rectifier with Smoothing Capacitor (AC to DC Converter)

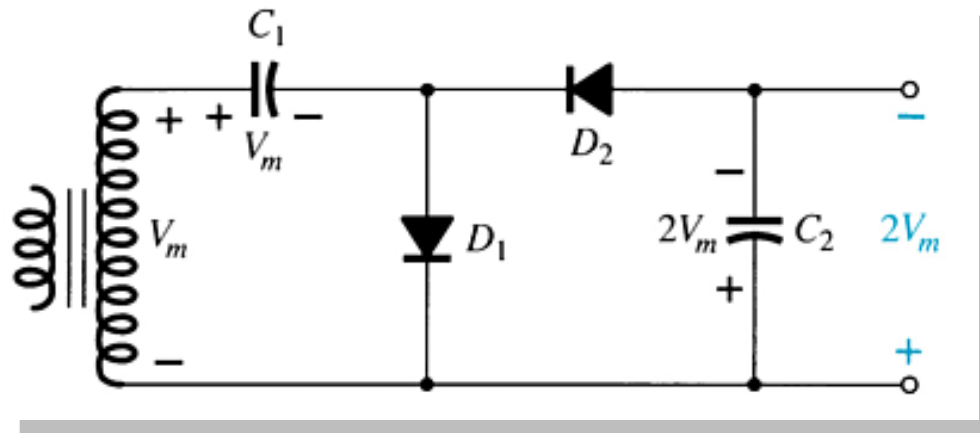


Voltage-Multiplier Circuits

Voltage multiplier circuits use a combination of diodes and capacitors to step up the output voltage of rectifier circuits.

- **Voltage Doubler**
- **Voltage Tripler**
- **Voltage Quadrupler**

Voltage Doubler



This half-wave voltage doubler's output can be calculated by:

$$V_{\text{out}} = V_{C2} = 2V_m$$

where V_m = peak secondary voltage of the transformer

$$-V_m - V_{C1} + V_{C2} = 0$$

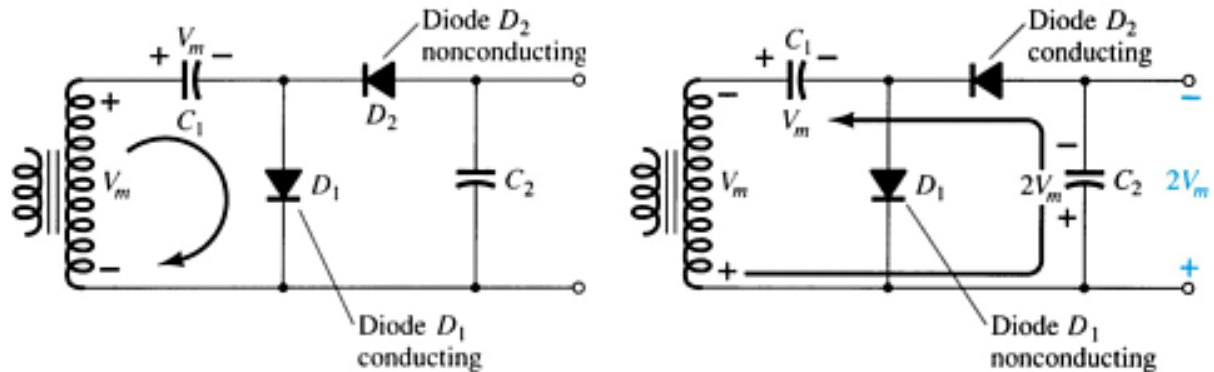
$$-V_m - V_m + V_{C2} = 0$$

$$V_{C2} = 2V_m$$

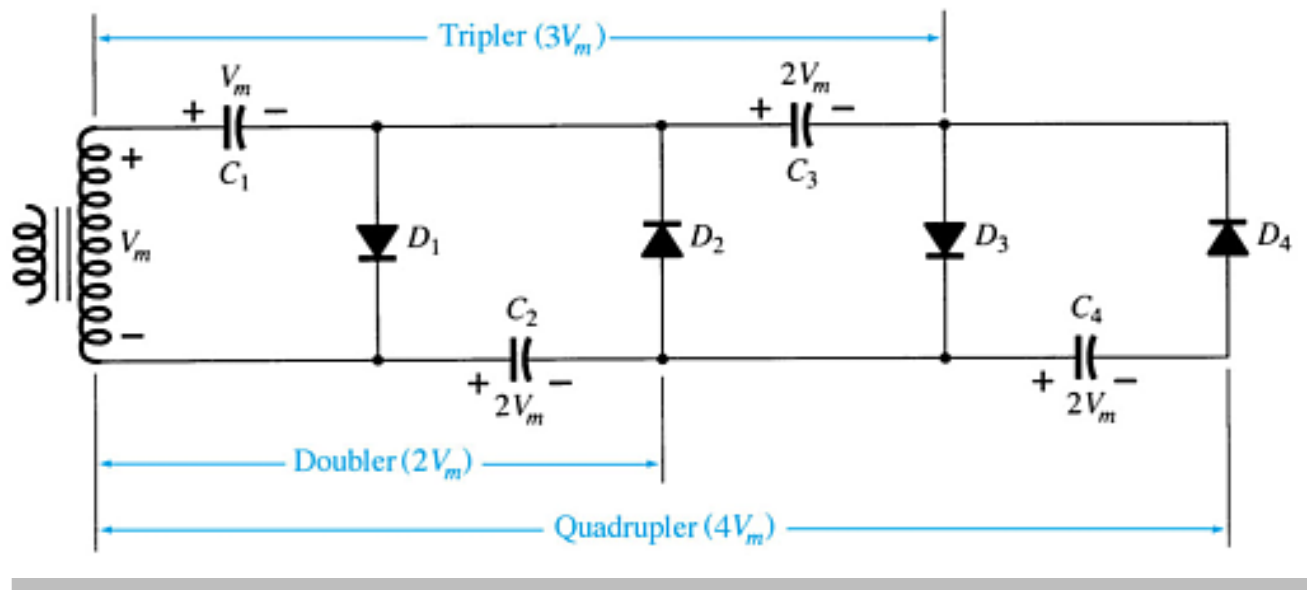
Voltage Doubler

- **Positive Half-Cycle**
 - D_1 conducts
 - D_2 is switched off
 - Capacitor C_1 charges to V_m
- **Negative Half-Cycle**
 - D_1 is switched off
 - D_2 conducts
 - Capacitor C_2 charges to $2V_m$

$$V_{\text{out}} = V_{C2} = 2V_m$$



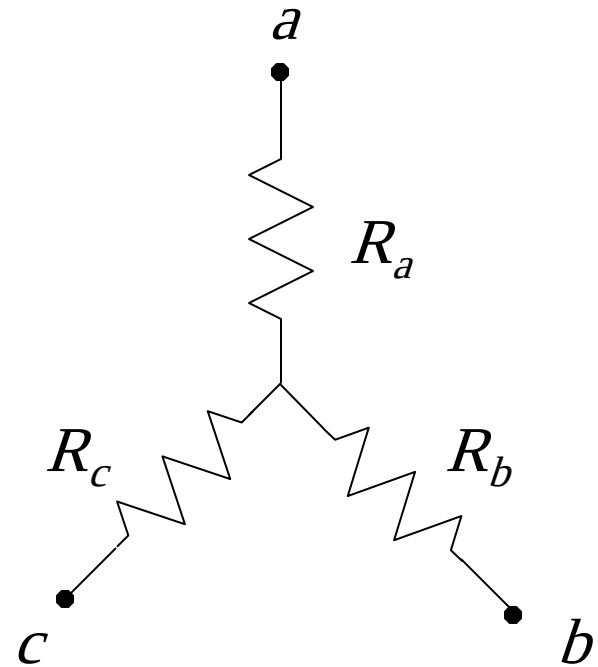
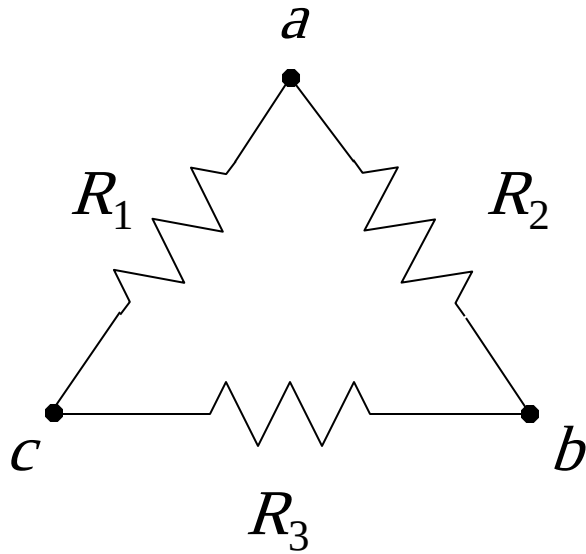
Voltage Tripler and Quadrupler



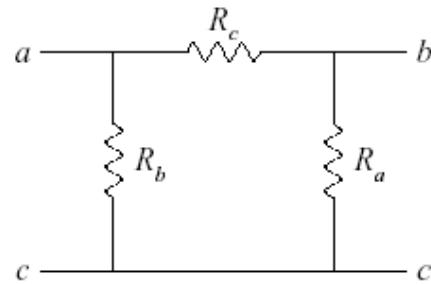
Δ -Y Transformation

- A particular configuration of resistors (or impedances) that does not lend itself to the using series and parallel combination techniques is that of a delta (Δ) connection
- In such cases the delta (Δ) connection is converted to a wye (Y) configuration
- The reverse transformation can also be performed

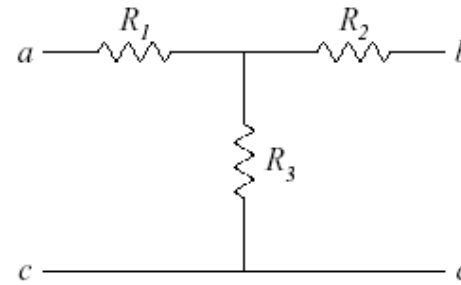
Δ -Y Transformation



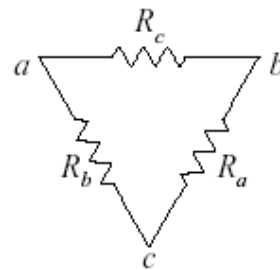
Delta Wye Transformations



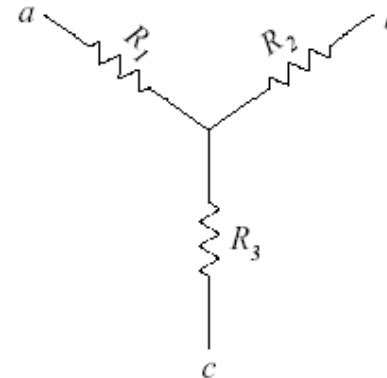
Delta



Wye

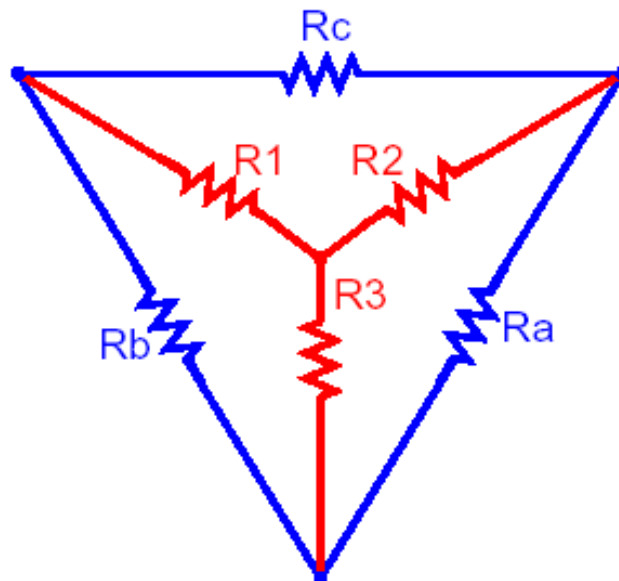


Delta



Wye

- Every Delta network is functionally equivalent to a Wye network (and vice versa).



The following must be satisfied for equivalence

$$R_a = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1}$$

$$R_b = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2}$$

$$R_c = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3}$$

$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c}$$

$$R_2 = \frac{R_c R_a}{R_a + R_b + R_c}$$

$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c}$$